5. Mortality

Contents

1 Environment preparation

Run scripts with the needed packages and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
```

2 Counts and cumulative proportions

2.1 By lineage, tumoral status, and sleep condition

```
df %>%
  group_by(lineage, tum_state, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 18 x 6
##    lineage tum_state sleep_condition   n_total n_dead  prop
##    <fct>   <fct>     <fct>               <int>  <dbl> <dbl>
```

```
##  1 HC_MT   Healthy   Control              48      1  2.08
##  2 HC_MT   Healthy   Sleep deprivation    48      6 12.5
##  3 HO_MT   Healthy   Control              62      1  1.61
##  4 HO_MT   Healthy   Sleep deprivation    38      1  2.63
##  5 HO_MT   Tumoral   Control              10      0  0
##  6 HO_MT   Tumoral   Sleep deprivation    34      1  2.94
##  7 HO_SPC  Healthy   Control              62      2  3.23
##  8 HO_SPC  Healthy   Sleep deprivation    29      1  3.45
##  9 HO_SPC  Tumoral   Control              10      0  0
## 10 HO_SPC  Tumoral   Sleep deprivation    43      0  0
## 11 HO_SPT  Healthy   Control              12      5 41.7
## 12 HO_SPT  Healthy   Sleep deprivation    25      3 12
## 13 HO_SPT  Tumoral   Control              60      2  3.33
## 14 HO_SPT  Tumoral   Sleep deprivation    47      0  0
## 15 HO_VLN  Healthy   Control              96      1  1.04
## 16 HO_VLN  Healthy   Sleep deprivation    96      0  0
## 17 HV_GAL  Healthy   Control              72      1  1.39
## 18 HV_GAL  Healthy   Sleep deprivation    72      6  8.33
```

2.2 By lineage and tumoral status

```
df %>%
  group_by(lineage, tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 9 x 5
##   lineage tum_state n_total n_dead  prop
##   <fct>   <fct>       <int>  <dbl> <dbl>
## 1 HC_MT   Healthy        96      7  7.29
## 2 HO_MT   Healthy       100      2  2
## 3 HO_MT   Tumoral        44      1  2.27
## 4 HO_SPC  Healthy        91      3  3.3
## 5 HO_SPC  Tumoral        53      0  0
## 6 HO_SPT  Healthy        37      8 21.6
## 7 HO_SPT  Tumoral       107      2  1.87
## 8 HO_VLN  Healthy       192      1  0.52
## 9 HV_GAL  Healthy       144      7  4.86
```

2.3 By lineage and sleep condition

```
df %>%
  group_by(lineage, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
```

```r
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 12 x 5
##    lineage sleep_condition   n_total n_dead  prop
##    <fct>   <fct>               <int>  <dbl> <dbl>
##  1 HC_MT   Control                48      1  2.08
##  2 HC_MT   Sleep deprivation      48      6 12.5
##  3 HO_MT   Control                72      1  1.39
##  4 HO_MT   Sleep deprivation      72      2  2.78
##  5 HO_SPC  Control                72      2  2.78
##  6 HO_SPC  Sleep deprivation      72      1  1.39
##  7 HO_SPT  Control                72      7  9.72
##  8 HO_SPT  Sleep deprivation      72      3  4.17
##  9 HO_VLN  Control                96      1  1.04
## 10 HO_VLN  Sleep deprivation      96      0  0
## 11 HV_GAL  Control                72      1  1.39
## 12 HV_GAL  Sleep deprivation      72      6  8.33
```

2.4 By tumoral status and sleep condition

```r
df %>%
  group_by(tum_state, sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 4 x 5
##   tum_state sleep_condition   n_total n_dead  prop
##   <fct>     <fct>               <int>  <dbl> <dbl>
## 1 Healthy   Control               352     11  3.12
## 2 Healthy   Sleep deprivation     308     17  5.52
## 3 Tumoral   Control                80      2  2.5
## 4 Tumoral   Sleep deprivation     124      1  0.81
```

2.5 By lineage

```r
df %>%
  group_by(lineage) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 6 x 4
```

```
##   lineage n_total n_dead  prop
##   <fct>     <int>  <dbl> <dbl>
## 1 HC_MT        96      7  7.29
## 2 HO_MT       144      3  2.08
## 3 HO_SPC      144      3  2.08
## 4 HO_SPT      144     10  6.94
## 5 HO_VLN      192      1  0.52
## 6 HV_GAL      144      7  4.86
```

2.6 By tumoral status

```
df %>%
  group_by(tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
##   tum_state n_total n_dead  prop
##   <fct>       <int>  <dbl> <dbl>
## 1 Healthy       660     28  4.24
## 2 Tumoral       204      3  1.47
```

2.7 By sleep condition

```
df %>%
  group_by(sleep_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
##   sleep_condition   n_total n_dead  prop
##   <fct>               <int>  <dbl> <dbl>
## 1 Control               432     13  3.01
## 2 Sleep deprivation     432     18  4.17
```